1. Outline the basic concept of Stevens' *psychophysical power law,* and why it is important in the design of information visualisation systems. In your answer, outline three different common styles of chart or graphical encoding, and critique each of them in terms of this power law.

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2. Multidimensional data is common in science and industry. As an example, here is an experimental data scheme from a biologist's spreadsheet, recording a series of measurements of rats' growth, with each rat being given one of four different foods during the experiment:

ID: integer; // a unique identifier for the measurement, e.g. 1234

dateOfTest: date; // the day the measurement was done, e.g. Jan 19, 2006

animal: integer; // the identifier for the rat, e.g. 2345

comment: string; // open text for lab notes, e.g. "Doesn't look too healthy, seems lethargic"

weight: real; // the mass of the rat in grams, e.g. 150.9

length: real; // the length in cm of the rat, e.g. 22.1

food: integer; // the identifier for the food given to the rat, e.g. 3

Using this data as an example, outline a system design for a visualisation centred on the *small multiples* design technique. In your answer, justify the way you categorise and graphically encode this data.

3. Imagine you work in part of a bank where a database of financial trades is key to many people's work. There are attributes such as the trade amount (in £), the yield %, the number of days until the bond is mature (e.g. 25), the customer ID (e.g. 12345), the currency (e.g. USD), the trade date, and the bank employee who did the trade (e.g. 2345). Workers use an SQL query system to examine old data, and to make reports, but the worker training takes a long time, and yet error rates in queries are high. Outline an alternative system, based on interactive filtering and querying, that is guaranteed to have no such SQL errors.